

**Stereotype
machines**

Workers

On the following pages, you will see grids of 36 images of 'workers' generated by artificial intelligence.

Which of the following grids of images do you think represents a group of Italian workers?

↗ **SKIP TO THE END**

Read how we generated the images with
Stable Diffusion

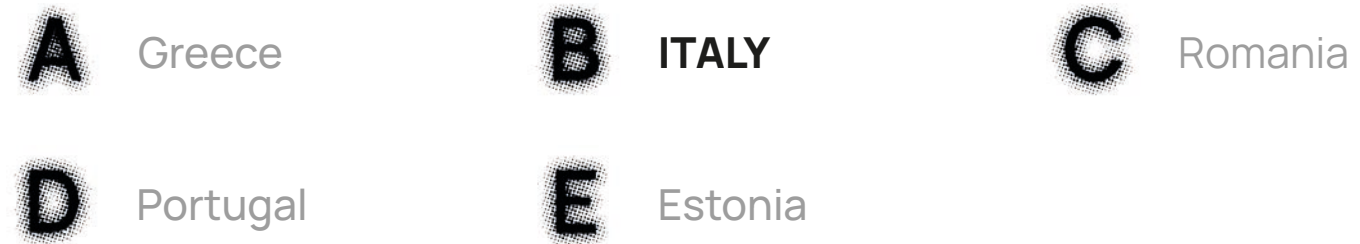












The AI-generated 'Italian workers' were grid B.

Was it easy for you to guess? Which visual elements indicated that these images represent people from that country?

Which visual stereotypes do the AI-generated images contain that might have clued you in?

→ **FLIP PAGE**

Dive deep into each nationality.

Italian workers

These are images of 'Italian workers' created by the text-to-image AI model Stable Diffusion.

As you can see, these AI-generated images share many common characteristics. How much do they truly represent Italian workers, and how much do they represent a stereotypical idea of them?

Look at the images together and ask yourself:

- How many people are in each group?
- What kind of industry does the image refer to?
- How much is it representative of Italian workers?

→ FLIP PAGE

Compare **objects**, **people** and **surroundings**.

↗ SKIP TO THE END

Read how we generated the images with Stable Diffusion.





ALIGN WITH THESE MARKS

- Are the images generated by the AI model an accurate representation of a group of Italian workers?
 - According to this AI model, what visual traits characterise an Italian worker?
 - What kind of jobs do these objects suggest?
- In this layer, you can see the objects generated by the AI model.



ALIGN WITH THESE MARKS

- Does the way they are dressed characterise the workers' nationality?
 - What do their clothes tell you about the represented jobs?
- In this layer, you can see the people generated by the AI model.

Italian workers

We have created layers from each of the images to allow you to analyse their components in more detail.

Turn the first page, and then read the **pink text** on the ← left.

Now turn to the second page, and read the **blue text**.

Focus on the surrounding context:

- **Are the workers portrayed indoors or outdoors?**
- **What kind of work is being done?**
- **Do these settings suggest typical 'workplaces'?**





ALIGN WITH THESE MARKS



ALIGN WITH THESE MARKS

Greek workers





ALIGN WITH THESE MARKS



ALIGN WITH THESE MARKS

Romanian workers





ALIGN WITH THESE MARKS



ALIGN WITH THESE MARKS

Portuguese workers





ALIGN WITH THESE MARKS



ALIGN WITH THESE MARKS

Estonian workers



Methodology

How we generated the images

From February to June 2024, we generated all the images in these books by feeding the same text prompt to 'Draw Things' (<https://drawthings.ai>), a free, open-source software that runs text-to-image generative AI models on your computer. Images were generated using a programmatic approach. Instead of using the user interface, we handled the software through code. The app is available for the latest generation of Mac computers. Specifically, we used M1/M3 Pro MacBooks.

As a base model, we used 'Stable Diffusion XL official model' (<https://huggingface.co/stabilityai/stable-diffusion-xl-base-1.0>), one of the most well-documented open-source text-to-image generative models, released in 2022.

It has been trained on a dataset indexing 5 billion images, called LAION-5B (<https://laion.ai/>). This dataset is one of the largest open-source collections for training AI models. Despite criticism for including some copyrighted or problematic images, it remains the most transparent and non-black-boxed resource available for AI model training.

To generate the images, we used the following prompt:

```
(A group of (****:1.5) people working), (35mm lens photography), extremely detailed, 4k, shot on dslr, photorealistic, photographic, sharp
```

Where **** changes by country in each collection of images: Estonian, Greek, Italian, Portuguese, Romanian.

We also used a negative prompt, stating what we didn't want to be generated. Some of these keywords are very technical,

but they are necessary to force the system to create images that could be part of a family album and to exclude images taken by security cameras, or with insufficient illumination, or images with non-credible human beings, to name some examples.

```
ng_deepnegative_v1_75t, (worst quality:2), (low quality:2), (normal quality:2), lowres, bad anatomy, normal quality, ((monochrome)), ((grayscale)), (verybadimagenegative_v1.3:0.8), negative_hand-neg, (lamp), badhandv4
```

However, the images you see in the book are not the direct result of the software, but rather a more elaborate protocol identified to create high-quality images.

The process from prompt to image involved the following steps:

- generating the initial image
- refining it based on the general composition
- refining complex parts such as faces, hands and objects
- increasing the image resolution
- fixing the details in the new high-resolution image

The described steps were sequentially applied for the generation of each image.

If you want to reproduce the images contained in this book, go to this link to access the source code:

<https://github.com/densitydesign/stereotype-machines-generator>